Investigation of additive manufactured catalysts for the methane oxidation reaction

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Introduction

Liquified natural gas is proposed as a possibility to reduce emissions in the maritime transport sector. Due to incomplete combustion of the methane in the engine, a further catalytic reduction is necessary. Here, monolithic catalysts are used. They consist of parallel straight channels and combine a high surface area with a low pressure drop. Since there is no connection between the channels, there is no mass transfer in the radial direction. Additive manufactured structures like POCS or woodpile monoliths are one possible development to overcome this. Both are strand-based structures and combine a high outer surface area with good radial mass transfer. This work presents a model for predicting the pressure drop of woodpile monoliths based on experimental data and the Ergun equation.

Experimental

Woodpile monoliths of aluminum oxide differing in macroporosity and outer surface area were manufactured using the direct ink writing technique. Compressed air and an acrylic tube with a diameter of 16 mm were used for pressure drop measurements. The monoliths were wrapped with Teflon tape to eliminate the potential of bypass flow. The outer surface area (a_v), and the macroporosity (ε_v) were calculated based on microscopic pictures. Starting with the Ergun equation[1] (1), a MATLAB code was developed to adjust the correction factors (α, β).

$$\frac{\Delta p}{\Delta l} = \alpha \cdot \frac{\eta_{air} \cdot a_v^2}{\varepsilon_v^3} \cdot u_{air,0} + \beta \cdot \frac{\rho_{air} \cdot a_v}{\varepsilon_v^3} \cdot u_{air,0}^2 \tag{1}$$

Here, $\frac{\Delta p}{\Delta l}$ is the pressure drop per bed height, η_{air} and ρ_{air} are the viscosity and the density of air, and $u_{air,0}$ is the velocity of the air in the tube.

Results

Figure 1 shows the experimentally measured pressure drop for a variation of the outer surface area (a)) and the macroporosity (b)). In the experiments, the outer surface area ranges from 4500 m^{-1} to 2700 m^{-1}, while the macroporosity is between 50% to 70%.

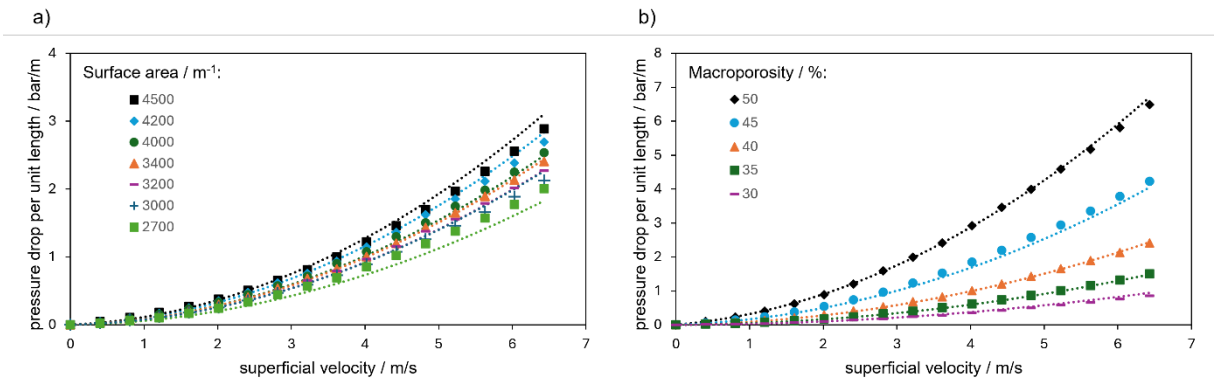

Figure 1: Pressure drop per length vs. superficial velocity where the points are the experimental data and the dotted lines are the calculated data according to equation (2). a) shows a variation in the outer surface area; b) is a variation in the macroporosity.

An increase in the outer surface area and decreasing macroporosity both increase the pressure drop. This is expected according to the Ergun equation. Fitting the parameters α and β equation (2) is obtained.

$$\frac{\Delta p}{\Delta l} = \frac{0.26}{\varepsilon_v^5} \cdot \frac{\eta_{air} \cdot a_v^2}{\varepsilon_v^3} \cdot u_{air,0} + \frac{0.18}{\varepsilon_v} \cdot \frac{\rho_{air} \cdot a_v}{\varepsilon_v^3} \cdot u_{air,0}^2 \qquad (2)$$

The calculated pressure drop curves with equation (2) are plotted in Figure 1 as dotted lines. The experimental data are well described by equation (2) in varying the outer surface area and over an extensive range of macroporosities. In contrast to Ergun's equation, the experiments show an even more significant influence of the macroporosity for the pressure drop of woodpile structures.

Conclusion and outlook

A pressure drop correlation based on Ergun's equation is obtained in this work. Comparing the correlation to experimental data, good accordance is found over a broad range of outer surface areas and macroporosities. Thus, a good predictability for the pressure drop of woodpile structures is expected and should be confirmed in future work.

Literature

[1] S. Ergun, A. A. Orning, *Industrial & Engineering Chemistry* **1949**, *41*, 1179-1184.